Big Data

BY THE END:

You can explain what makes data big and know when ordinary tools are no longer enough.

Overview

A small furniture shop may record about 20 sales per day in one table, while a digital wallet may record millions of payments per day. Both store transaction data, but the digital wallet needs tools that can handle much more data arriving much faster.

What makes data big

Big data is data that is too large, arrives too quickly, or comes in too many formats for ordinary tools to handle well.

There is no single number that makes data big. What matters is whether the usual tools can still store and process it.

The three Vs

PART	SIMPLE QUESTION	SMALL FURNITURE SHOP	DIGITAL WALLET
Volume	How much data is there?	About 20 sales per day	Millions of payments per day
Velocity	How quickly does new data arrive?	A few sales per hour	Thousands of payments per second
Variety	How many formats does it use?	One sales table	Payment records, app logs, messages, and receipt photos

A dataset can be difficult to handle because of one V or several at the same time.

Big depends on the tools

Two organizations can use different tools for the same dataset. A file with 2 million rows may be too large for a spreadsheet but easy for a database.

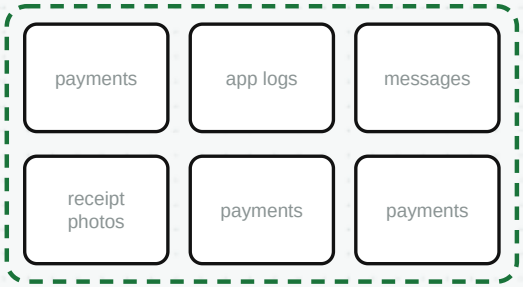
Data becomes big when the tools normally used are no longer enough. Many small shops, schools, and clinics can answer their questions with ordinary tools. They do not need big data systems.

A small furniture shop



about 20 sales per day
a few sales per hour

A digital wallet



millions of payments per day
thousands of payments per second

What changes when data is big

When one machine is not enough, the work is divided across several machines.

WHAT CHANGES	WHAT IT MEANS
Storage	Data may be stored across several machines
Processing	Several machines may work on the data together
Quality	Missing or incorrect values are harder to find among millions of records
Management	Copies of data are harder to track, protect, and delete

More data is not always better

More data is not always better. A small, clean dataset can be more useful than a large dataset with missing or incorrect values.

The furniture shop can answer a useful question with 20 accurate sales records. Millions of unreliable records may give the wrong answer.